

equal probability. Namely, equities will either earn 8% (the average), earn 35%, or they will lose 19% in any given year. The arithmetic average of these three numbers is exactly 8%; this cycle is illustrated in Figure 6.6. Some of you may have seen this type of handy triangle before.

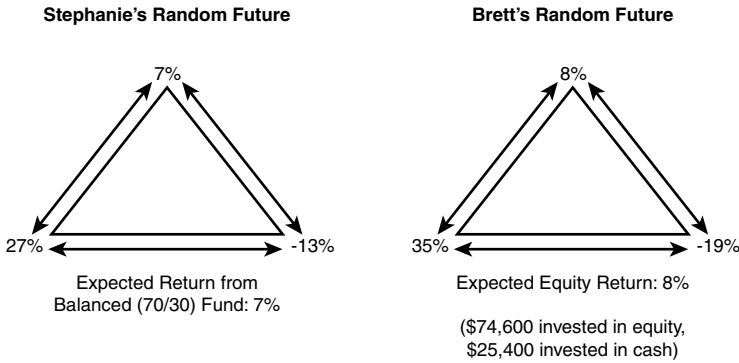


Figure 6.6 Who will be better off?

Source: M. Milevsky, "Spending Buckets and Financial Placebos," *Research Magazine*, June 2007.

The standard deviation of this variable consisting of three possible investment returns is the square root of the expression: $(1/3)(0) + (1/3)(0.27)^2 + (1/3)(0.27)^2$, which is approximately 21.9%.

By virtue of the fact that Stephanie has invested in a fund that has 70% in "triangular equities" and 30% in cash (and to be consistent with our previous assumptions), Stephanie's fund will also earn one of three possible investment returns. She will either earn 27% (the good) or she will lose 13% (the bad) or she will earn 7% (the average). All of them have an equal probability of occurring. Note that the standard deviation for her fund's random return is the square root of $(1/3)(0) + (1/3)(0.20)^2 + (1/3)(0.20)^2$, which is 16.3%. And, just to convince yourself that the math works out, notice that $(\$74,600/\$100,000) \times 21.9\%$ is also 16.3%. In other words, at the point of retirement both of them have an equivalent total asset allocation but a very different strategic plan for how to generate an income during the next three years. Table 6.3 illustrates the possible returns that Stephanie and Brett will encounter.